

**Behavioral and Academic Lifestyle Analysis for Navigating College Equilibrium
(BALANCE): Modeling the Student Balance Index using Educational Data Mining**



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CHAPTER I

INTRODUCTION

The increasing digitalization of higher education has normalized technology-integrated and hybrid learning as a permanent feature of university systems worldwide. Students now navigate heavy and continuous academic demands, mediated entirely through screens, which are associated with high levels of perceived stress and burnout [1], [2], [3]. Burnout in students, typically characterized by emotional exhaustion, cynicism, and reduced academic efficacy, is consistently linked to impaired mental health, lower academic performance, and higher dropout risk [2], [4], [5].

Globally, screen time and "digital burnout" have emerged as critical stressors. Extensive engagement with digital platforms, multiple concurrent online courses, and sustained cognitive load are key predictors of poorer psychological health and greater academic procrastination [6], [7], [8]. Parallel evidence indicates that unhealthy lifestyle habits common in student populations, such as insufficient sleep and low physical activity, co-occur with high perceived stress and are inversely related to study engagement [9]. Together, heavy workloads and unbalanced daily routines form a cluster of modern educational stressors that universally drive burnout risk [4], [9].

In the Philippines, where rapid digitalization coexists with structural constraints. Filipino university students exhibit very high levels of internet and social media use, which shows a double-edged association with mental health [10]. While internet use can foster support, it is simultaneously linked to greater psychological distress when engagement becomes excessive [10]. Furthermore, digital inequalities, such as relying on unstable internet connections or lacking personal devices, exacerbate stress, causing students from resource-constrained households to experience significantly higher anxiety [11], [12].

At the local level, such as in rapidly developing urban centers like Davao City, these infrastructural challenges are compounded by physical logistical burdens. Commonly reported challenges include crowded home environments and the need to manage exhausting commutes in dense urban areas when attending hybrid sessions [13], [11]. These localized conditions intensify cognitive fatigue. Because university-based mental health services in these regional institutions are often resource-constrained, systematic and proactive screening for burnout is incredibly difficult to implement at scale [13].

To address this, Educational Data Mining (EDM) has become a prominent tool to identify at-risk students using machine learning [14], [15]. However, a critical research gap exists: current EDM applications are predominantly reactive and performance-centric [16]. Predictors are largely lagging indicators, such as midterm grades or accumulated LMS activity, which only signal problems

once academic performance has already deteriorated [17], [15]. To fill this gap, this study proposes **Project BALANCE**, developing a proactive Student Balance Index (SBI). By synthesizing daily lifestyle data, including sleep, physical activity, and screen time, into a continuous, actionable metric, this project aims to enable earlier, context-sensitive interventions before severe burnout occurs.

REVIEW OF RELATED LITERATURE

Student Balance Index

The Student Balance Index (SBI) is a multidimensional composite metric designed to quantify the equilibrium between a student's academic commitments and their physiological and mental recovery. Transitioning from traditional, reactive GPA-centric models, the SBI operationalizes student success as a function of input sustainability, such as sleep, exercise, and diet, rather than purely through output performance. By mathematically synthesizing these behavioral features using Educational Data Mining (EDM) techniques, the SBI functions as a proactive diagnostic tool. It identifies "at-risk" behavioral profiles where high academic effort may be leading toward burnout, or where external burdens destabilize a student's academic equilibrium.

The parameters selected for this index represent a "Modern Student Stressor Cluster," validated by contemporary educational research. Sleep Hours and Exercise Frequency are included as primary recovery mechanisms, as insufficient sleep and sedentary behavior are proven leading predictors of emotional exhaustion and diminished cognitive efficacy in university cohorts [18]. Conversely, Study Hours and Screen Time (comprising social media and entertainment usage) represent the cumulative cognitive load; research indicates that excessive digital engagement, when not balanced by recovery, correlates significantly with "digital burnout" and academic procrastination [19].

Educational Data Mining

Recent EDM and educational data science work increasingly extend beyond purely academic variables by incorporating behavioral and lifestyle attributes (e.g., study habits, screen exposure, health-related behaviors) as input features. Large reviews of predictive models in education note a gradual shift toward integrating non-traditional variables—motivation, habits, and socio-behavioral factors—alongside grades and demographics to improve prediction quality [20][21]. In multi-factor student performance datasets, behavioral variables such as study hours, reading frequency,

transportation mode, and engagement time are collected through questionnaires and platform logs, then encoded using label encoding for ordinal variables and one-hot encoding for nominal variables, followed by normalization or standardization to ensure comparable scales [22]. Feature importance and selection procedures (e.g., Chi-square tests, Recursive Feature Elimination, ensemble-based ranking) are used to identify the most influential behavioral predictors and reduce dimensionality [19][22][23].

Although most EDM studies still emphasize academic and engagement features, some adjacent work on mental health and productivity explicitly models lifestyle composites. In survey-based ML studies on students' mental health and productivity, variables such as sleep duration, screen time, physical activity, diet, and stress scores are jointly analyzed, with feature importance analysis (e.g., in Random Forest or logistic models) revealing that sleep patterns, stress scores, and screen time are among the strongest predictors of academic productivity and wellbeing [24]. In burnout research more broadly, explainable ML studies combine psychometric scales and work-life indicators into composite burnout scores and then apply feature selection methods (e.g., Boruta) to determine which demand-resource variables (e.g., psychological distress, organizational support) contribute most to the overall burnout index[25][26]. These approaches illustrate methodological patterns relevant to constructing a Student Balance Index: standardized collection of daily behavioral measures, transformation into numeric features, and weighting through data-driven feature importance rather than ad hoc weighting schemes [22][24][26].

Unsupervised learning, particularly K-Means clustering, is widely used in EDM to discover latent student profiles from behavioral and performance data. K-Means is commonly applied after preprocessing steps such as handling missing values, feature scaling (e.g., min-max or z-score normalization), and sometimes feature selection to reduce noise[30][31][32]. The number of clusters is typically determined using internal validity indices such as the elbow method and silhouette analysis, which evaluate within-cluster cohesion and between-cluster separation [27][33][29]. Some studies further refine K-Means by introducing statistical criteria to objectively select the optimal K, improving clustering stability and downstream prediction reliability 27[25].

Clustering is often performed on behavioral engagement metrics (e.g., study hours, attendance, tutoring sessions, LMS clickstream patterns) to derive interpretable profiles such as “high-achiever, highly engaged,” “average performer,” and “at-risk, low engagement” students [30] [31] [32]. For example, in higher education datasets, K-Means has been used to generate three to five clusters that distinguish students by combinations of attendance, study effort, and learning outcomes; high-achieving clusters tend to show consistently high study hours and attendance, while at-risk clusters show low engagement across behavioral indicators[28] [29] [31] In some hybrid frameworks,

cluster labels are fed into supervised models (e.g., SVM) as additional features or as refined class labels, yielding higher predictive accuracy than using raw grades alone [28] [27]. Beyond performance, clustering has been applied in burnout research: K-Means-based segmentation of healthcare trainees identified high-, intermediate-, and low-burnout clusters, with profiles characterized by work-life imbalance and demographic patterns [26]. These methods demonstrate how K-Means can support behavior-based student profiling, which is directly relevant to deriving discrete “balance profiles” underlying a Student Balance Index.

Supervised machine learning is the dominant approach in EDM for predicting academic performance, dropout, and, more recently, wellbeing-related risks, using academic, demographic, and behavioral features. Comparative studies consistently show that ensemble methods (Random Forest, Gradient Boosting, XGBoost, LightGBM, CatBoost) often outperform traditional models like single Decision Trees or basic Logistic Regression, especially on large, heterogeneous educational datasets [19] [22] [35] [36] [37]. In multi-algorithm benchmarks, Gradient Boosting or Random Forest frequently achieve the highest accuracy (often >80%), F1-scores, and ROC-AUC values for predicting final grades or pass/fail outcomes, with feature importance analyses highlighting early assessments and engagement metrics as key predictors [21] [19] [22].

At the same time, simpler models such as Logistic Regression, SVM, and Decision Trees remain widely used due to interpretability and robustness. Several EDM studies report logistic regression or SVM matching or slightly trailing deep models in accuracy (e.g., ~80–86%), while offering clearer coefficients or decision rules for educators [20] [38] [22] [36] [37]. Deep learning architectures (e.g., deep neural networks) have been applied to VLE clickstream data, using handcrafted behavioral features and time-series inputs; these models can reach 84–93% accuracy and outperform logistic regression and SVM in detecting at-risk students early in a course [38] [36].

In burnout-specific prediction, supervised ML has been leveraged in occupational and caregiver domains, where Decision Trees, Random Forests, and other classifiers trained on psychometric scales and lifestyle indicators successfully classify burnout levels, with tree-based models highlighting key drivers such as exhaustion, work–life balance, and financial stress [25] [26]. A survey of ML in burnout prediction suggests that the choice of algorithm depends on data structure and outcome type, but ensemble tree methods and multitask learning frameworks generally provide strong performance while handling complex, multi-factor burnout constructs [25] [26]. Collectively, these findings indicate that tree-based ensembles and, in some contexts, deep learning are well suited for predictive modeling over behavioral datasets, and that integrating engineered lifestyle features into such models is a promising direction for early detection of student burnout and imbalance.

EDUCATIONAL PROBLEM

The core educational problem facing modern higher education institutions is the inability to proactively monitor and manage student burnout in a digitally integrated learning environment. While existing EDM interventions demonstrate that predictive models can be integrated into institutional workflows to inform targeted support, they rarely embed multi-dimensional metrics of student daily well-being [14], [17]. Currently, risk is usually operationalized purely in terms of course failure or withdrawal. The predictors used by institutions are largely lagging indicators such as poor grades, lack of participation logs, and end-of-unit assessments which only flag a student for intervention once their academic performance has irreparably deteriorated [17].

This performance-centric approach completely ignores the root causes of academic decline. In a system characterized by high digital engagement and limited mental health resources, such as Philippine universities, there is a clear and urgent need to shift the analytic focus from retrospective academic failure to the ongoing equilibrium between academic demands and personal health. Without a holistic, composite index to measure this balance, administrators and counselors are unable to identify emerging behavioral imbalances before they manifest as severe academic decline.

STATEMENT OF THE PROBLEM

This study aims to address the reliance on reactive academic metrics by developing and validating a leading indicator, the Student Balance Index (SBI), utilizing Educational Data Mining (EDM) techniques. Specifically, this research seeks to answer the following questions:

1. How can diverse, non-academic lifestyle variables (sleep hours, exercise frequency, daily study hours, screen time, and stress levels) be mathematically synthesized to formulate a standardized Student Balance Index (SBI)?
2. What hidden behavioral clusters or student profiles emerge when grouping students based on their SBI components?
3. To what extent does a low Student Balance Index score accurately predict negative educational outcomes, specifically high dropout risk and lower exam scores?

RESEARCH OBJECTIVES

To develop, cluster, and validate a Student Balance Index (SBI) using educational data mining techniques to provide university support teams with a proactive tool for identifying academic burnout risk. The specific objectives are the following:

1. To develop a composite Student Balance Index (SBI) by assigning appropriate weights to behavioral and lifestyle features (sleep, exercise, study hours, screen time, and stress) within the dataset.
2. To identify distinct student lifestyle profiles by applying unsupervised clustering techniques (e.g., K-Means) to the components of the SBI, allowing for targeted intervention strategies.
3. To evaluate the predictive validity of the SBI by employing supervised machine learning models to test its correlation with and ability to predict target academic outcomes (dropout risk and exam scores).

SIGNIFICANCE OF THE STUDY

This study holds significant potential for the following stakeholders:

University Administrators and Student Support Services. University administrators, guidance counselors, and academic advisors will benefit from this study by gaining a data-driven framework for identifying early signs of student imbalance and burnout risk. The proposed Student Balance Index (SBI) provides an interpretable metric that integrates behavioral and lifestyle indicators into a single measure of student equilibrium. By using the SBI as a proactive monitoring tool, institutions can better prioritize support services, design targeted interventions, and implement early-warning systems that help prevent academic decline and student dropout.

Students and the Academic Community. Students will benefit from insights generated by the BALANCE framework by gaining a clearer understanding of how lifestyle behaviors influence academic performance and wellbeing. The identification of distinct lifestyle patterns and behavioral clusters can inform awareness campaigns and wellness initiatives that encourage healthier study routines, balanced digital engagement, and improved time management. In turn, this can foster a more supportive academic environment that promotes both academic success and sustainable student wellbeing.

Educational Data Mining Researchers and Education Technology Developers. This study contributes to the advancement of Educational Data Mining by introducing the Student Balance Index (SBI) as a novel metric that integrates non-academic behavioral variables into predictive educational analytics. Researchers and developers may build upon this framework to explore improved modeling techniques, incorporate additional behavioral or contextual data sources, or design intelligent decision-support systems for educational institutions. The findings may therefore serve as a foundation for future research on wellbeing-oriented learning analytics and proactive student support systems.

SCOPE AND DELIMITATIONS

This study focuses on the development and validation of a Student Balance Index (SBI) using Educational Data Mining (EDM) techniques to examine the relationship between student lifestyle behaviors and academic outcomes. The research utilizes a secondary dataset containing behavioral, lifestyle, and academic-related variables of university students, including sleep hours, exercise frequency, daily study hours, screen time, and stress level. These variables will be mathematically synthesized to construct the Student Balance Index as a measure of students' academic-life equilibrium. The study applies unsupervised clustering methods, particularly K-Means clustering, to identify distinct student lifestyle profiles based on SBI components, and supervised machine learning models to evaluate the predictive validity of the SBI in relation to selected academic outcomes, specifically dropout risk and exam scores.

This research is limited to the variables available in the secondary dataset and does not include additional psychological, environmental, or institutional factors that may also influence student burnout or academic performance. The Student Balance Index is constructed only from selected behavioral indicators which represent a simplified model of student balance rather than a comprehensive measure of student wellbeing. Furthermore, the predictive models developed in this study are intended for analytical and exploratory purposes only and are not designed to function as a fully operational early-warning system for universities. As such, the findings should be interpreted as an initial framework for understanding student balance and informing future research on wellbeing-oriented educational data mining.

CHAPTER II

METHODOLOGY

DATA COLLECTION

The dataset utilized in this study was obtained from the publicly available Student Habits and Academic Performance Dataset hosted on the Kaggle platform. The dataset was published by Aryan208 and is accessible through the Kaggle repository for educational and machine learning

research purposes. Public repositories such as Kaggle provide open datasets that enable researchers to explore relationships between behavioral patterns and academic outcomes using data-driven approaches. The dataset contains structured information describing students' demographic characteristics, lifestyle behaviors, psychological indicators, and academic outcomes. These variables collectively capture aspects of students' daily routines, including study habits, digital media consumption, physical health behaviors, and mental well-being. Such variables are commonly used in educational analytics to investigate how lifestyle and behavioral factors influence student academic performance and overall wellbeing.

The dataset consists of multiple records representing individual students and includes a wide range of features relevant to student behavior and academic performance. Key variables include demographic attributes (e.g., age, gender, and major), behavioral habits (e.g., study hours per day, social media usage, Netflix consumption, sleep duration, and exercise frequency), lifestyle indicators (e.g., diet quality and screen time), psychological variables (e.g., stress level, motivation level, and exam anxiety score), and contextual factors such as parental education level, family income range, and internet quality. In addition, the dataset contains academic outcome variables including attendance percentage, previous grade point average, exam score, and dropout risk.

These variables allow for the analysis of how student lifestyle behaviors and psychological factors interact with academic outcomes. The dataset is particularly suitable for Educational Data Mining (EDM) because it integrates both academic and non-academic attributes, enabling the development of predictive and behavioral models related to student success and wellbeing.

Since the dataset is publicly available and contains anonymized information, no personally identifiable data is included. Consequently, the dataset complies with ethical standards for secondary data analysis. The data was downloaded in comma-separated values (CSV) format and subsequently imported into a data analysis environment for preprocessing, feature engineering, and modeling.

The selected dataset provides a comprehensive representation of student lifestyle patterns and academic performance indicators, making it appropriate for developing the proposed Student Balance Index (SBI) and for applying machine learning techniques to analyze student behavioral profiles and predict academic outcomes.

DATA PROCESSING

After collection, the dataset underwent several preprocessing steps to ensure that it was suitable for analysis and machine learning modeling. These steps included data cleaning, feature preparation, normalization, and the computation of the proposed **Student Balance Index (SBI)**.

First, the dataset was inspected for missing values, duplicate entries, and inconsistent records. Duplicate rows were removed to avoid bias in the analysis. Missing values in numerical variables were handled using median imputation, while categorical variables were filled using the mode. This ensured that the dataset remained complete without significantly distorting the distribution of the variables. Next, categorical attributes such as gender, major, learning style, and family income range were transformed into numerical representations using encoding techniques. Nominal variables were converted using one-hot encoding, while ordinal variables were encoded using label encoding to preserve their ranked relationships. This transformation allows the variables to be processed by machine learning algorithms.

Because the variables in the dataset exist on different measurement scales (e.g., hours, percentages, and rating scores), numerical features were normalized to ensure comparability across variables.

Standardization using z-score normalization was applied so that each feature has a mean of zero and a standard deviation of one:

$$Z = \frac{X - \mu}{\sigma}$$

In this equation, X represents the original feature value, μ represents the mean of the feature, and σ represents its standard deviation. The resulting standardized value Z ensures that variables with larger numeric ranges do not dominate the model.

Following normalization, the **Student Balance Index (SBI)** was computed as a composite metric representing the equilibrium between student recovery behaviors and academic or digital strain. The index is constructed from selected lifestyle variables that reflect students' daily routines and psychological state.

The recovery component is computed as a weighted combination of normalized recovery variables:

$$Recovery = 0.4Z_{sleep} + 0.3Z_{exercise} + 0.3Z_{diet}$$

The strain component represents activities and conditions that contribute to academic pressure or burnout risk. It is calculated using normalized study load, screen exposure, and stress level:

$$Strain = 0.3Z_{study} + 0.3Z_{screen} + 0.4Z_{stress}$$

The **Student Balance Index** is obtained by subtracting the strain score from the recovery score:

$$SBI = Recovery - Strain$$

Because this raw calculation utilizes Z-scores, the resulting values can include negative numbers and decimals. While this raw, unscaled SBI serves as the mathematically optimal central feature for the machine learning algorithms (clustering and predictive modeling), it is less intuitive for human interpretation. Therefore, for reporting purposes and dashboard visualization, a final Min-Max scaling step is applied to transform the raw SBI into a standardized 0 to 100 range:

$$SBI_{scaled} = \frac{SBI - \min(SBI)}{\max(SBI) - \min(SBI)} \times 100$$

Within this scaled framework, a higher score (approaching 100) indicates a highly resilient, well-balanced student, while a score approaching 0 flags severe imbalance and high burnout risk. This dual approach ensures computational accuracy for the algorithms while providing an actionable, highly interpretable metric for educational stakeholders.

PROPOSED DATA MINING TECHNIQUES

This study applies a combination of unsupervised and supervised machine learning techniques to analyze student lifestyle patterns and evaluate the effectiveness of the proposed Student Balance Index (SBI). These techniques allow the study to both discover hidden behavioral patterns among students and assess the predictive capability of the SBI with respect to academic outcomes.

First, K-Means clustering will be employed as an unsupervised learning technique to identify latent behavioral profiles among students based on their lifestyle and balance-related attributes. The clustering process will primarily utilize the components used in constructing the Student Balance Index, such as sleep hours, exercise frequency, study hours per day, screen time, and stress level. K-Means partitions the dataset into clusters by minimizing the distance between data points and their

assigned cluster centroids, allowing students with similar lifestyle patterns to be grouped together. The optimal number of clusters will be determined using the Elbow Method and Silhouette Analysis, which evaluate clustering compactness and separation. Through this clustering process, the study aims to uncover meaningful student profiles, such as students with balanced routines, students experiencing digital overload, or students with high academic strain and low recovery behaviors. These clusters will provide insight into the different lifestyle patterns that may influence academic performance and wellbeing.

Second, supervised machine learning models will be used to evaluate the predictive validity of the computed SBI, as it will be used as a key feature in predicting academic outcomes, particularly exam scores and dropout risk. For classification tasks involving dropout risk, algorithms such as Logistic Regression, Decision Trees, and Random Forest may be applied to classify students into risk categories. For regression tasks involving exam score prediction, models such as Linear Regression or Random Forest Regression will be used to estimate academic performance based on lifestyle and behavioral variables.

These supervised models will help determine whether the Student Balance Index is a meaningful indicator of student success and whether lower SBI values are associated with higher academic risk. Feature importance analysis from tree-based models will also be examined to identify which lifestyle factors contribute most significantly to student balance and academic outcomes.

By combining clustering analysis and predictive modeling, the study leverages Educational Data Mining techniques to both explore behavioral student profiles and validate the predictive usefulness of the Student Balance Index. This dual approach allows the research to move beyond simple correlation analysis and instead provide deeper insights into how lifestyle behaviors influence academic performance and burnout risk.

EVALUATION CRITERIA

To assess the effectiveness of the proposed models and the validity of the Student Balance Index (SBI), several evaluation metrics will be employed for both clustering and predictive modeling tasks. These metrics will measure the quality of student grouping as well as the performance of machine learning models in predicting academic outcomes.

For the clustering analysis, internal cluster validation metrics will be used to determine the optimal number of clusters and evaluate the quality of the generated student profiles. The Elbow Method will be applied to analyze the within-cluster sum of squares (WCSS) and identify the point

where adding more clusters no longer significantly improves clustering performance. Additionally, Silhouette Analysis will be used to measure how similar each data point is to its assigned cluster compared to other clusters. Higher silhouette scores indicate better-defined and more separated clusters.

For the supervised learning models, different evaluation metrics will be applied depending on the prediction task. For classification tasks, particularly for predicting dropout risk, model performance will be evaluated using Accuracy, Precision, Recall, and F1-score. Accuracy measures the proportion of correct predictions made by the model, while precision and recall evaluate how effectively the model identifies students at risk. The F1-score provides a balanced measure of precision and recall, making it particularly useful when evaluating classification models with potential class imbalances.

For regression tasks, such as predicting exam scores, the models will be evaluated using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). These metrics measure the difference between predicted and actual values, allowing the study to determine how accurately the models estimate student academic performance.

Additionally, the relationship between the computed Student Balance Index (SBI) and academic outcomes will be examined through model performance and correlation analysis. If lower SBI values consistently correspond to higher dropout risk or lower exam scores, this will indicate that the proposed index is a meaningful indicator of student balance and academic wellbeing.

Together, these evaluation criteria provide a comprehensive framework for assessing both the quality of the clustering results and the predictive performance of the machine learning models, ensuring that the findings of the study are reliable and scientifically valid.

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